

Lecture 06 - Professor Solution: Measuring Risk, Return, and Empirical Dependence

Instructor Overview

This mini-case is designed as an integration point after the lectures on descriptive statistics, probability, random variables, distributions, and the normal model. It uses one financial dataset to connect several ideas that students often learn separately: transforming prices into returns, summarizing an empirical distribution, comparing empirical evidence with a probability model, and interpreting conditional probabilities as evidence about dependence.

The updated case uses daily Meta Platforms adjusted closing prices from 2018 through July 10, 2026. The statistical target is not to forecast Meta Platforms or recommend a trade. The target is to teach students how to move from raw financial data to a careful historical risk-return statement using a current, unfamiliar-enough dataset.

Data and Setup

The dataset is:

Lecture 06 - meta daily 2018 2026.csv

The correct price column is Adj Close. Students should use adjusted close prices because they account for corporate actions and therefore give a more appropriate return series.

Simple daily returns are computed as:

$$R_t = (P_t - P_{(t-1)}) / P_{(t-1)}.$$

The first return is missing because a one-period return requires two prices. With 2141 price observations, the return sample contains 2140 daily returns.

Item | Value

Asset | Meta Platforms

Ticker | META

Price observations | 2141

Return observations | 2140

First return date | 2018-01-03

Last return date | 2026-07-10

Price column | Adj Close

Return definition | Simple daily return

The price series is useful as context, but the statistical analysis should focus on returns rather than prices. Prices are levels; returns are the object for risk-return analysis.



Historical Risk-Return Summary

The daily return summary is:

Statistic | Value

Mean | 0.0960%

Median | 0.0999%

Standard deviation | 2.6177%

Standard error of the mean | 0.0566%

Minimum | -26.3901%

1st percentile | -6.4365%

5th percentile | -3.7370%

95th percentile | 3.6718%

99th percentile | 7.2272%

Maximum | 23.2824%

Share of negative days | 47.9439%

Skewness | -0.2785

Excess kurtosis | 17.0279

The sample average daily return is positive, but it is small relative to the daily standard deviation. This is a key finance lesson: the mean return and volatility live on very different scales at the daily frequency. The standard error of the mean, about 0.0566%, is useful because it reminds students that the sample mean is an estimate, not the true expected return.

Annualized volatility is:

$s_{\text{textannual}}$ approx. $0.026177\sqrt{252}=0.4156$.

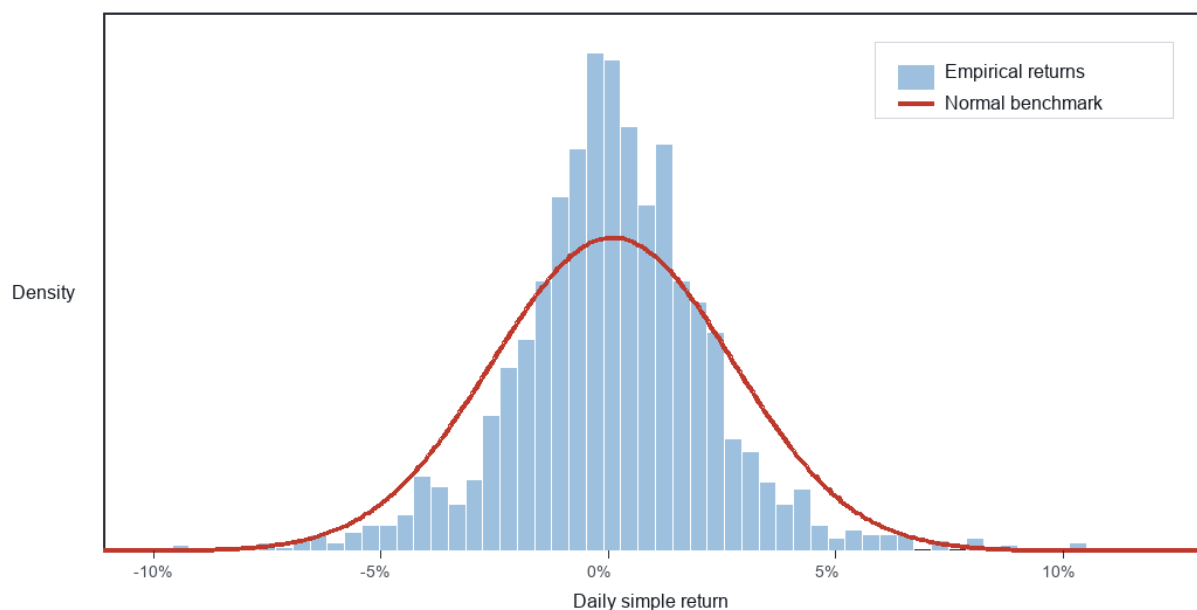
So the annualized volatility is approximately 41.56%. This should be described as a scale conversion from daily volatility to annual volatility, not as a guaranteed future annual risk level.

A simple annualization of the mean using $((1+\text{mean } R)^{252}-1)$ gives approximately 27.37%. This number is useful pedagogically but should be interpreted with great caution. The mean is estimated with uncertainty, and average returns are much less stable than volatility estimates.

Empirical Distribution and Normal Benchmark

The histogram of daily returns is centered close to zero, but the distribution has visible extreme observations. The excess kurtosis is 17.0279, which is far above the normal benchmark of zero. This means the empirical distribution has substantially heavier tails than a normal distribution.

META daily returns: empirical histogram and normal benchmark



Using the sample mean and standard deviation, the normal-benchmark 5th percentile is:

$\text{mean } R - 1.645s = -4.2101\%$.

The empirical 5th percentile is -3.7370%. At the 5% level, the normal benchmark is not identical, but it is in the same broad range as the empirical cutoff.

The lower tail becomes more revealing at more extreme thresholds:

Tail event | Empirical frequency | Normal benchmark

$(R_t < \text{mean } R - 2s) \mid 1.9626\% \mid \text{about } 2.28\%$

$(R_t < \text{mean } R - 3s) \mid 0.6075\% \mid \text{about } 0.135\%$

The event below (mean $R-2s$) is higher than the normal benchmark, and the event below (mean $R-3s$) is much more frequent than a normal model would suggest. This is an important conclusion: the normal model can be a useful central benchmark while still understating extreme tail risk.

Independence Analysis: Ordinary Loss Days

To connect with the probability lecture, define:

$$L_t = R_t < 0.$$

This is the event that day (t) is a loss day. The relevant comparison is between:

$$P(L_t)$$

and

$$P(L_t \mid L_{(t-1)}).$$

If loss days are independent from the previous day's loss indicator, these probabilities should be approximately equal.

Using the 2139 observations with a valid lagged return event:

Quantity | Value

$(P(L_t)) \mid 47.9663\%$

$(P(L_t \mid L_{(t-1)})) \mid 47.0760\%$

$(P(L_t \mid \text{not } L_{(t-1)})) \mid 48.7871\%$

The contingency table is:

Previous day | Current non-loss | Current loss

Previous non-loss | 570 | 543

Previous loss | 543 | 483

For ordinary loss days, the conditional probability after a previous loss is slightly lower than the unconditional probability. The difference is small. The clean teaching interpretation is that this sample does not show strong evidence that ordinary loss days cluster from one day to the next.

Students should not overstate this. The conclusion is not "returns are independent." It is narrower: this particular binary event, in this historical sample, does not show strong dependence using this simple one-day comparison.

Independence Analysis: Large-Loss Days

Now define a large loss as a return at or below the empirical 5th percentile:

$$B_t = R_t \leq q_{(0.05)}.$$

Here:

$$q_{(0.05)} = -3.7370\%.$$

The empirical comparison is:

Quantity | Value

$(P(B_t))$ | 5.0023%

$(P(B_t | B_{(t-1)}))$ | 7.4766%

$(P(B_t | \text{not } B_{(t-1)}))$ | 4.8720%

The contingency table is:

Previous day | Current non-large-loss | Current large loss

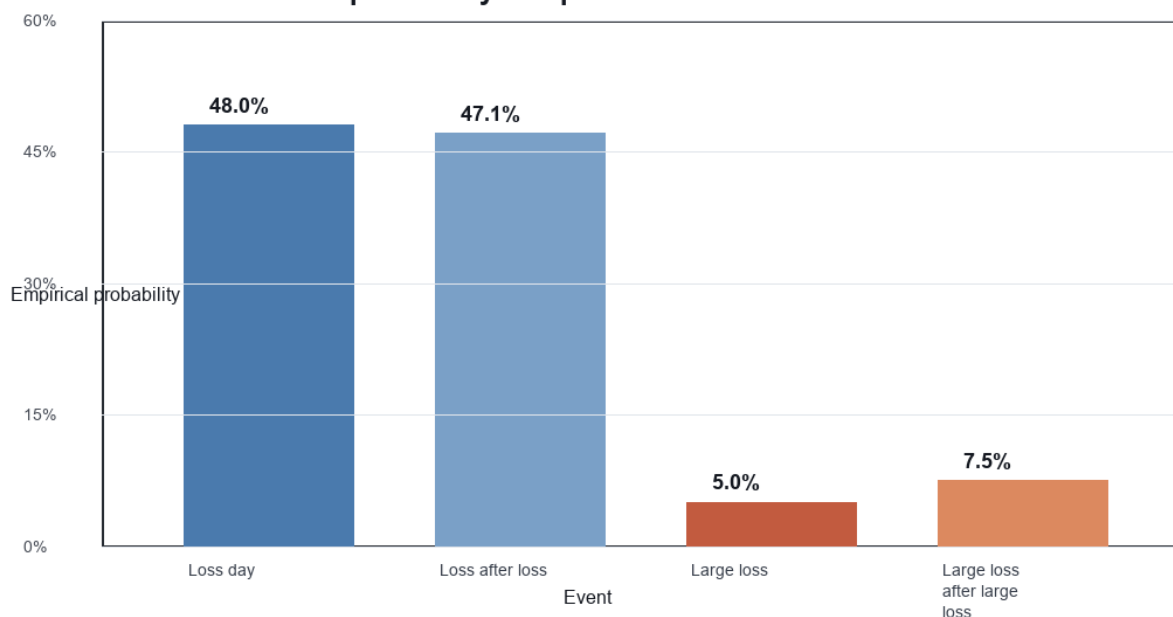
Previous non-large-loss | 1933 | 99

Previous large loss | 99 | 8

This part gives a richer discussion. The unconditional probability of a large loss is about 5.0%, by construction. Conditional on the previous day also being a large loss, the probability rises to about 7.5%. That suggests some evidence of clustering in large-loss days.

The sample-size caution is essential: there are only 107 days on which the previous day was a large-loss day, and only 8 cases of consecutive large-loss days. The result is suggestive, not definitive.

META conditional probability comparison



Numeric Self-Check Answer Key

This compact key can be separated from the full solution if instructors want students to check their calculations before seeing the memo.

Quantity | Answer

Price observations | 2141

Return observations | 2140

First return date | 2018-01-03

Last return date | 2026-07-10

Mean daily return | 0.0960%

Daily standard deviation | 2.6177%

Standard error of the mean | 0.0566%

Empirical 5th percentile | -3.7370%

Normal-benchmark 5th percentile | -4.2101%

$(P(L_t))$ | 47.9663%

$(P(L_t | L_{(t-1)}))$ | 47.0760%

$(P(B_t))$ | 5.0023%

$(P(B_t | B_{(t-1)}))$ | 7.4766%

Suggested Final Memo

Using adjusted close prices from January 2018 to July 2026, I computed simple daily returns for Meta Platforms and summarized the resulting historical return sample. The sample contains 2140 daily returns. The average daily return was positive at about 0.096%, while the daily standard deviation was about 2.62%. This means the average

return is small relative to day-to-day volatility, so the return distribution cannot be understood from the mean alone.

The annualized volatility estimate is approximately 41.56%, using the square-root-of-time conversion. This should be read as a scale conversion from historical daily volatility, not as a fixed prediction of future annual risk. Downside observations were material: the worst daily return was about -26.39%, and the empirical 5th percentile was about -3.74%.

The normal benchmark is useful as a reference point, but the empirical evidence points to heavier tails. In particular, returns below three standard deviations from the mean occurred more frequently than the normal model would suggest. The independence analysis gives a nuanced result. Ordinary loss days do not appear to cluster strongly in this one-day comparison, since the probability of a loss after a previous loss is close to the unconditional loss probability. Large-loss days, however, appear somewhat more likely after a previous large-loss day, although the number of such cases is small.

Overall, Meta Platforms had a positive average return in this historical sample, but the evidence also shows meaningful volatility, downside risk, heavy tails, and some possible dependence in extreme losses. These are historical sample findings, not guarantees about future performance.